Automated GRD Prediction from Clinical Data Using Machine Learning: A Multi-Class Classification Approach for Hospital El Pino

Cristopher Pradenas, Matías Alé, and Diego Lobos Díaz Emiliano Fernandez Felipe salas ortiz

***Abstract*—The automated assignment of *Grupos Relacionados por el Diagnóstico* (GRD) codes is indispensable for hospital resource planning and healthcare financing under Chile's Fondo Nacional de Salud (FONASA) reimbursement framework. Manual coding, however, remains labour-intensive and is inherently susceptible to inter-coder variability. We propose a supervised machine learning framework to automate this process using structured clinical data from Hospital El Pino, comprising 14,561 inpatient episodes and 526 unique GRD categories. Our pipeline applies binary TF-IDF encoding to CIE-10 diagnostic and CIE-9 procedural fields, yielding a 2,995-dimensional feature matrix with 99.28 % sparsity. Four architectures—Logistic Regression, Random Forest, XGBoost, and a Multilayer Perceptron (MLP)—were systematically evaluated. The MLP demonstrates superior discriminative capability, achieving an accuracy of 0.8168 and a macro-averaged F1-score of 0.7433 on the 80 most frequent GRD classes, which collectively account for 70.8 % of the dataset. These results establish the viability of automated, structured-data pipelines as low-latency decision-support tools for clinical coding in Chilean public hospitals.**

***Index Terms*—GRD prediction, diagnosis-related groups, clinical coding automation, multi-class classification, multilayer perceptron, CIE-10, CIE-9, Hospital El Pino, high-dimensional sparsity, binary TF-IDF encoding.**

I. Introduction

The *Grupos Relacionados por el Diagnóstico* (GRD) system constitutes a patient classification framework adapted in Chile from the Diagnosis-Related Groups (DRG) methodology developed in the United States [1]. FONASA employs GRD codes to determine hospital reimbursement rates under the *Modalidad de Atención Institucional* [2], rendering accurate code assignment a direct determinant of hospital revenue and resource allocation. In current clinical practice, coding specialists retrospectively assign GRD labels by reviewing principal and secondary diagnoses, procedures, and discharge summaries—a process that is labour-intensive and generates measurable inter-coder variability [3]. Automating this retrospective process through machine learning has consequently emerged as a critical research priority for reducing administrative overhead in healthcare systems [4], [5].

This study frames automated GRD prediction as a supervised multi-class classification problem defined on structured patient-level data: CIE-10 diagnostic codes, CIE-9 procedural codes, patient age, and biological sex. The primary computational challenges are three-fold: 526 unique GRD categories distributed across an extreme imbalance ratio of 813:1; a feature space exhibiting 99.28 % sparsity; and the unordered, multi-valued nature of the diagnostic and procedural fields. To address these challenges, we evaluate four architectures—ranging from tree-based ensembles such as XGBoost [6] to feedforward neural networks—and assess their capacity to extract discriminative clinical patterns from this high-dimensional tabular representation [7], [8].

The study is conducted on 14,561 inpatient records from Hospital El Pino, Santiago, Chile, each episode described by up to 35 CIE-10 diagnostic codes and up to 30 CIE-9 procedural codes. To ensure full reproducibility, the preprocessing pipeline, model implementations, and trained artefacts are publicly available at: **[https://github.com/Cristxpherr/CINF104-GRD-HospitalElPino]**. The remainder of this paper is structured as follows: Section II surveys related work; Section III formulates the study objectives; Section IV details the methodology; Section V presents the exploratory data analysis; Section VI reports experiments, results, and discussion; Sections VII and VIII state conclusions and limitations; and Section IX describes data and code availability.

II. Related Work

Building on Slee's foundational patient classification framework [1], the automation of clinical coding has evolved from natural language processing (NLP) applied to free-text discharge summaries [4], [5] to predictive modelling on structured electronic health records (EHRs). This shift is primarily motivated by the quantifiable economic impact of manual coding inconsistencies on hospital revenue cycles [3].

For structured, tabular clinical data, gradient-boosted tree ensembles such as XGBoost [6] constitute highly competitive baselines, benefiting from native support for sparse feature matrices and built-in L_1/L_2 regularisation. Nevertheless, when input representations attain extreme dimensionality and sparsity—as is the case with multi-hot ICD code encodings—feedforward neural architectures have demonstrated superior capacity to distil latent clinical representations that transcend the expressive limits of tree-based splitting criteria [7].

In the domain of automated DRG prediction, Janke et al. [9] reported macro F1-scores of 0.70–0.82 on curated subsets of US hospital data using logistic regression and random forests, while Boag et al. [10] established the strong predictive

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value of clinical code co-occurrence patterns from MIMIC-III. Li et al. [11] proposed graph-based ICD code embeddings leveraging the hierarchical ICD taxonomy, achieving accuracy above 0.85 on curated MIMIC-III subsets. In marked contrast, predictive GRD modelling in Latin America remains scarce; regional literature is predominantly oriented toward administrative cost analysis [2]. This study addresses that gap by applying structured machine learning to a real-world Chilean hospital cohort, navigating the compounded challenges of a large-cardinality, severely imbalanced GRD label space and 99.28 % feature sparsity.

III. Objective of the Study

The primary objective is to develop and rigorously evaluate a supervised machine learning model capable of automatically predicting the GRD code for a patient hospitalization episode from structured clinical data available at the time of discharge. The following specific sub-objectives guide the investigation:

(i) To conduct a thorough exploratory data analysis (EDA) of the Hospital El Pino cohort, characterising data quality, class distribution, and discriminative feature properties.

(ii) To design and implement a preprocessing and feature engineering pipeline that transforms sparse, multi-valued CIE-10 and CIE-9 code fields into a form amenable to standard classification algorithms.

(iii) To train and compare multiple supervised learning architectures—Logistic Regression, Random Forest, XGBoost, and a Multilayer Perceptron—on the prediction task.

(iv) To select the best-performing model according to macro-averaged classification metrics appropriate for the severely imbalanced label distribution.

(v) To contextualise the obtained results within the broader DRG/GRD prediction literature.

IV. Methodology

A. Dataset Description

The dataset comprises 14,561 retrospective inpatient episodes from Hospital El Pino, Santiago, Chile. Demographic variables include biological sex—female patients constitute a majority of 66.0 % (n=9,617) versus 34.0 % male (n=4,944)—and age (mean = 39.43 years, SD=24.68, median = 36, IQR = 23–60). The clinical trajectory for each episode is encoded via one principal and up to 34 secondary *CIE-10* diagnosis codes, alongside one principal and up to 29 secondary *CIE-9* procedure codes. Secondary fields display a missingness rate of 81.8 % for diagnoses and 57.9 % for procedures—values that reflect intrinsic variation in clinical complexity rather than data entry deficiencies. The target variable is the GRD code, encompassing 526 unique categories with a class imbalance ratio of 813:1. All records were de-identified prior to analysis in compliance with patient privacy regulations.

B. Development Method

The pipeline initiates with rigorous data cleaning and text normalisation. Each compound code field of the form `CODE-Description` is parsed to retain only the alphanumeric prefix (e.g., `A41.8` from `A41.8-Otras septicemias`). Missing age values are imputed with the training-set median. To ensure stratified splitting validity, 76 singleton GRD classes are removed, yielding a final analytical corpus of **14,485** records across **450** active GRD classes. A stratified 80/20 partition produces 11,588 training records and 2,897 hold-out test records, preserving the imbalanced clinical distribution across both subsets.

C. Machine Learning Techniques

Feature engineering transforms CIE-10 diagnostic codes and CIE-9 procedural codes into a unified numerical representation via binary TF-IDF vectorisation with $\mathrm{min_df}=2$. The set of all principal and secondary CIE-10 diagnosis codes per episode is concatenated as a bag of tokens; an identical operation is applied to CIE-9 procedure codes. The two resulting binary matrices (2,365 diagnostic features; 628 procedural features) are horizontally concatenated with z-score-standardised age and binary-encoded sex, producing the final **2,995-dimensional** feature matrix (sparsity: **99.28 %**).

Four supervised learning architectures are evaluated:

- **Logistic Regression (LR):** Multinomial classifier with L_2 regularisation (C=1.0, *saga* solver), linear baseline suited to sparse binary matrices.
- **Random Forest (RF):** Bagging ensemble of 100 decision trees ($\mathrm{max_depth}=20$, $\mathrm{min_samples_leaf}=2$).
- **XGBoost:** Regularised gradient-boosted ensemble (80 estimators, $\mathrm{max_depth}=5$, η=0.15, subsample = 0.8) [6].
- **Multilayer Perceptron (MLP):** Two hidden layers (256 and 128 neurons), ReLU activations, L_2 regularisation ($\alpha=10^{-4}$), Adam optimiser ($\eta=10^{-3}$), batch size 512, and early stopping (patience = 10) on a 10 % internal validation split.

D. Evaluation Metrics

The class imbalance ratio of 813:1 renders overall accuracy an inadequate primary metric, as it disproportionately rewards correct classification of majority classes while masking degraded performance on rare but clinically significant GRDs. Accordingly, the **macro-averaged F1-score**—which computes F1 independently per class and averages without frequency weighting—is adopted as the principal optimisation criterion. Weighted F1-score and overall accuracy are additionally reported. A normalised confusion matrix restricted to the top-12 GRD categories is analysed to identify systematic misclassification patterns.

V. Exploratory Data Analysis

A. Data Quality Assessment

A systematic assessment establishes the dataset's integrity prior to modelling. As illustrated in Fig. 1, both the principal CIE-10 diagnosis field and the principal CIE-9 procedure field exhibit 0.0 % missingness, providing a complete clinical baseline for every episode. Secondary fields display a

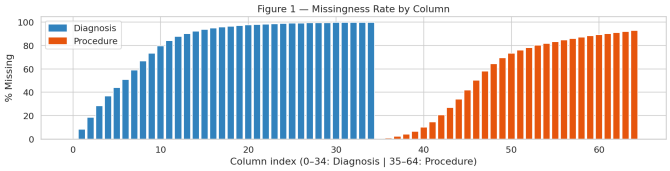

Fig. 1. Data completeness profile by column index. Both the principal CIE-10 diagnosis field (index 0) and the principal CIE-9 procedure field (index 35) exhibit 0.0 % missingness. Secondary fields display monotonically increasing sparsity—a signature of clinical coding practice rather than data entry error. **Blue**: CIE-10 diagnosis columns (indices 0–34); **orange**: CIE-9 procedure columns (indices 35–64).

TABLE I
SUMMARY STATISTICS OF THE HOSPITAL EL PINO COHORT

Variable	Value
Total hospitalization records	14,561
Unique GRD codes	526
Records after singleton removal	14,485
Active GRD classes	450
Age: mean ± SD	39.4 ± 24.7 years
Age: median (IQR)	36 years (23–60)
Age: range	0–121 years
Female patients	9,617 (66.0 %)
Male patients	4,944 (34.0 %)
Singleton GRDs excluded	76
Class imbalance ratio	813:1
Most frequent GRD (146101)	813 records (5.6 %)
Top-10 GRDs cumulative coverage	27.8 %
Top-20 GRDs cumulative coverage	40.3 %
GRDs required to cover 80 %	118 categories

monotonically increasing missingness rate with column index, averaging 81.8 % for CIE-10 secondary diagnoses and 57.9 % for CIE-9 secondary procedures—an intrinsic characteristic of clinical coding practice that validates the sparse vectorisation strategy adopted in Section IV-C.

B. Descriptive Statistics

Table I summarises the primary demographic and clinical characteristics of the cohort. After the exclusion of 76 singleton GRD categories, the active label space comprised 450 classes. The severity of class imbalance is underscored by the dominant category *PH CESÁREA* (146101), which alone accounts for 813 records (5.6 %), while 118 categories are required to cumulatively cover 80 % of all records.

Fig. 2 depicts the 20 most frequent GRD categories by record count, confirming the dominance of obstetric and neonatal GRDs in the hospital's casemix and directly motivating macro-averaged evaluation criteria.

C. Visual Analysis

Fig. 3 presents the cohort age distribution stratified by biological sex. Female patients exhibit a pronounced concentration in the 20–35 year range, directly corresponding to the obstetric

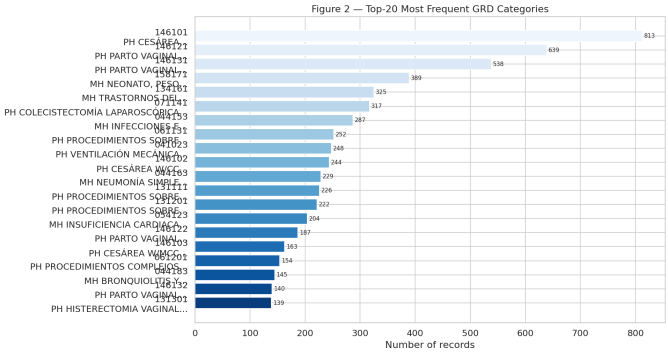

Fig. 2. Frequency distribution of the 20 most prevalent GRD categories. The casemix is dominated by obstetric codes (Cesárea, Parto Vaginal, Trastornos del Anteparto) and neonatal codes, establishing the clinical rationale for adopting macro-averaged metrics to preclude systematic bias toward majority classes.

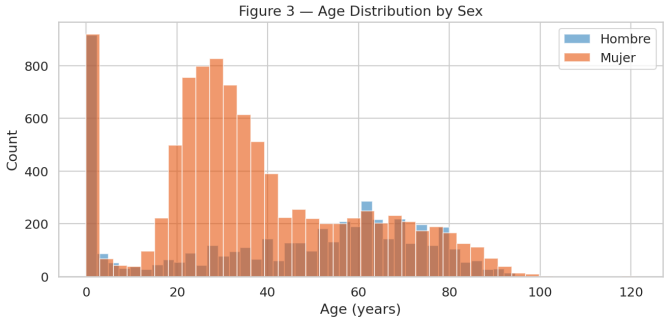

Fig. 3. Age distribution stratified by sex. The prominent female concentration in the 20–35 year range confirms the strong obstetric component of the Hospital El Pino casemix. Male patients manifest a broader, older distribution consistent with a chronic-disease admission profile.

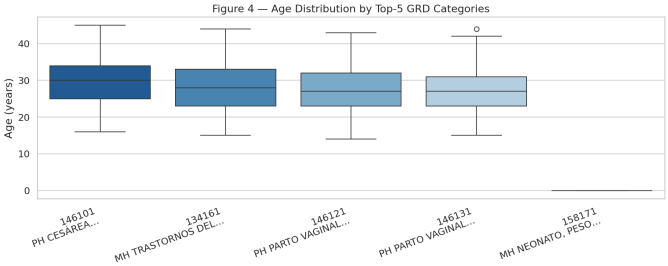

Fig. 4. Age distributions across the five most frequent GRD categories. Distinct, minimally overlapping interquartile ranges confirm that patient age functions as a highly discriminative predictor, with the Neonato category (age ≈ 0) exhibiting the most extreme separation.

GRDs identified in Fig. 2. Male patients display a broader, right-shifted distribution, reflecting greater prevalence of chronic and acute medical conditions.

Fig. 4 presents age boxplots stratified by the five most frequent GRD categories. Well-separated medians and narrow interquartile ranges across groups establish patient age as a strongly discriminative predictor. The Neonato GRD localises tightly around age zero; obstetric GRDs concentrate between 20 and 35 years; and acute medical GRDs skew toward older adult populations.

A Pareto analysis of GRD label frequencies is shown in Fig. 5. The cumulative frequency curve crosses the 80 %

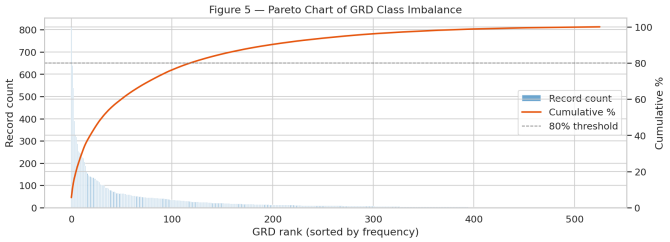

Fig. 5. Pareto chart of GRD label frequency. The cumulative frequency curve (red line) crosses the 80 % threshold (dashed line) at approximately 118 GRD categories, demonstrating the extreme long-tail distribution of the label space and directly motivating macro-averaged evaluation over frequency-weighted alternatives.

TABLE II
MODEL PERFORMANCE ON THE FULL 450-CLASS TEST SET (2,897 RECORDS)

Model	Acc.	Macro F1	Wtd. F1	Time (s)
Logistic Regression	0.6072	0.2054	0.5478	31
Random Forest	0.4550	0.0868	0.3509	1
XGBoost	0.5906	0.2356	0.5679	386
MLP	**0.6769**	**0.3497**	**0.6643**	**26**

threshold at only 118 of the 526 GRD codes, leaving a substantial long tail of rare categories. This structure constitutes the primary statistical justification for prioritising the macro-averaged F1-score over overall accuracy.

VI. EXPERIMENTS, RESULTS, AND DISCUSSION

A. Feature Engineering and Encoding

Input features were derived by applying binary TF-IDF vectorisation to the concatenated sets of all principal and secondary CIE-10 diagnosis codes, yielding a diagnostic vocabulary of 2,365 tokens, and independently to all CIE-9 procedure codes, yielding a procedural vocabulary of 628 tokens. The two binary matrices were horizontally concatenated with z-score-standardised age and binary-encoded sex, producing the final **2,995-dimensional** sparse feature matrix (sparsity: **99.28 %**). The GRD target variable was mapped to contiguous integer indices via a `LabelEncoder` fitted exclusively on the training partition to prevent information leakage.

B. Model Training and Comparison

All four candidate architectures were trained on the 80 % stratified partition (11,588 records) and evaluated on the held-out test set (2,897 records). Table II reports performance across the complete 450-class label space, where the severe class imbalance depresses absolute macro metrics. Table III restricts evaluation to the top-80 most frequent GRD categories (70.8 % of the dataset; 2,058 test records) to provide a granular characterisation of discriminative capability. Fig. 6 presents the visual comparison.

On the top-80 subset, the MLP attains the highest macro F1-score (0.7433) and accuracy (0.8168). Logistic Regression establishes a competitive linear baseline (macro F1 = 0.6854), consistent with the effective high-dimensional separability

TABLE III
MODEL PERFORMANCE ON THE TOP-80 GRD CATEGORIES (70.8 % OF DATASET; 2,058 TEST RECORDS)

Model	Acc.	Macro F1	Macro P	Macro R
Logistic Regression	0.7823	0.6854	0.7643	0.6633
Random Forest	0.6171	0.4072	0.5423	0.4074
XGBoost	0.7726	0.6722	0.7191	0.6546
MLP ⋆	**0.8168**	**0.7433**	**0.7637**	**0.7356**

⋆ Selected best model. P = Precision; R = Recall. All metrics macro-averaged.

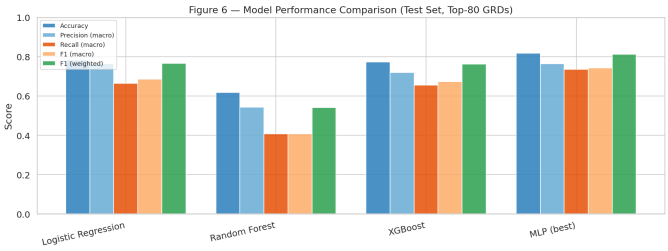

Fig. 6. Comparative performance of all four architectures on the top-80 GRD categories (held-out test set). The MLP consistently attains the highest scores across accuracy and macro-averaged precision, recall, F1-score, and weighted F1-score. XGBoost and Logistic Regression rank second and third, respectively; Random Forest exhibits the weakest performance, consistent with the limitations of bagging ensembles on extremely sparse binary feature spaces.

of sparse CIE-10 and CIE-9 code patterns [5]. XGBoost, constrained to 80 estimators by computational resources, achieves macro F1 = 0.6722; a fully tuned configuration is expected to reduce this gap substantially. Random Forest exhibits the weakest macro F1 (0.4072), reflecting the well-documented limitation of bagging ensembles on sparse binary matrices where individual feature contributions are diffuse.

C. Best Model Selection

The MLP was selected as the definitive architecture on the basis of its superior performance across *both* the full 450-class label space (macro F1 = 0.3497, accuracy = 0.6769) and the top-80 GRD subset (macro F1 = 0.7433, accuracy = 0.8168). The MLP also demonstrates remarkable operational efficiency, requiring only 26 s for training compared to 386 s for XGBoost. Macro F1 improvements over competing architectures on the top-80 subset are: +5.8 pp over Logistic Regression (0.7433 vs. 0.6854), +7.1 pp over XGBoost (vs. 0.6722), and +33.6 pp over Random Forest (vs. 0.4072).

D. Final Model Architecture

The selected MLP ingests the 2,995-dimensional sparse feature vector and propagates it through a feedforward network comprising two fully connected hidden layers of 256 and 128 neurons, each followed by a ReLU activation function and L_2 regularisation (α=10^{-4}). The output layer consists of a Softmax-activated projection over the 80 target classes (top-80 evaluation), optimised via categorical cross-entropy with the Adam algorithm (η=10^{-3}, batch size = 512). Training employs early stopping with patience = 10 monitored on a stratified 10 %

TABLE IV
MLP ARCHITECTURE AND TRAINING CONFIGURATION

Component	Specification	Output Dim.
Input	Sparse 2,995-dim. vector	2,995
Hidden layer 1	Dense + ReLU + L_2	256
Hidden layer 2	Dense + ReLU + L_2	128
Output layer	Dense + Softmax	80
Optimiser	Adam (η=10^{-3})	—
Loss function	Categorical cross-entropy	—
Regularisation	L_2, α=10^{-4}	—
Batch size	512	—
Early stopping	Patience = 10 (val. F1)	—

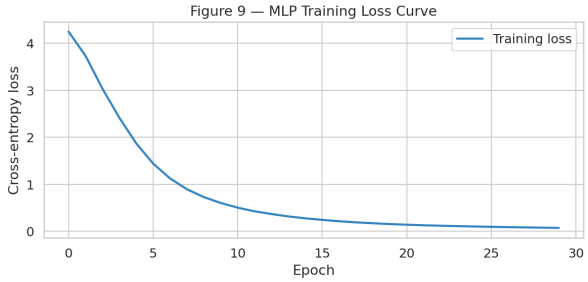

Fig. 7. Training cross-entropy loss curve of the MLP across epochs. The monotonically decreasing trajectory stabilises within 20–25 epochs before early stopping is triggered, confirming successful optimisation convergence and the absence of overfitting on the sparse clinical feature matrix.

internal validation split. The complete architecture is specified in Table IV.

E. Training Dynamics

Fig. 7 presents the MLP's training cross-entropy loss trajectory. The monotonically decreasing curve stabilises within 20–25 epochs, confirming that the Adam optimiser successfully minimises the objective function without gradient instability or oscillatory divergence. Smooth convergence coupled with the effectiveness of early stopping validates the network's capacity to generalise discriminative patterns from the high-dimensional sparse feature space.

F. Performance Analysis

Fig. 8 presents the normalised per-class recall confusion matrix for the top-12 most frequent GRD categories on the held-out test set. The high density along the principal diagonal—with multiple classes attaining recall = 1.00— demonstrates the MLP's discriminative precision for obstetric, neonatal, and elective surgical GRDs, whose clinical profiles are sharply characterised by age, sex, and CIE-9 procedure code signatures. Off-diagonal confusion is most pronounced between GRD pairs sharing a principal CIE-10 diagnosis but differing only in complication or comorbidity severity (MCC vs. CC)—a distinction that hinges on the presence of specific sparse secondary CIE-10 codes, and that constitutes the most challenging sub-problem even for experienced human coders.

Fig. 9 presents the top-20 most discriminative features ranked by mean absolute Logistic Regression coefficient

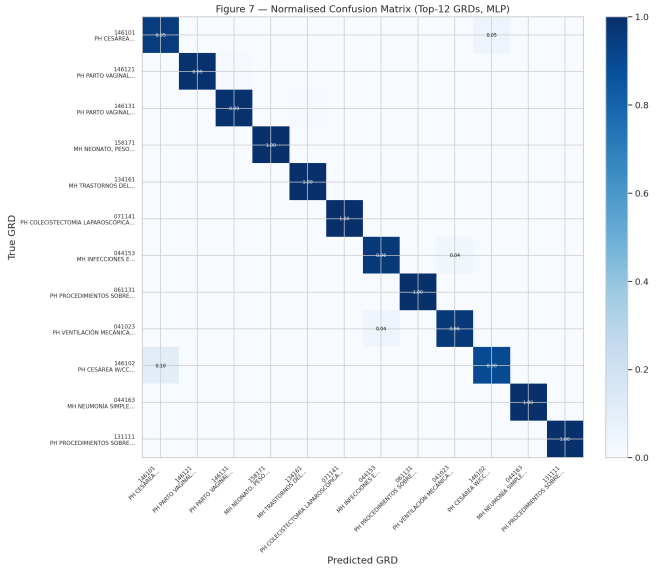

Fig. 8. Normalised per-class recall confusion matrix for the top-12 GRD categories (MLP, held-out test set). Diagonal values represent per-class recall. Near-unity diagonal entries for obstetric and neonatal GRDs reflect strong demographic and procedural separability, while off-diagonal values reveal the inherent difficulty of severity-level differentiation between related CIE-10 diagnostic categories.

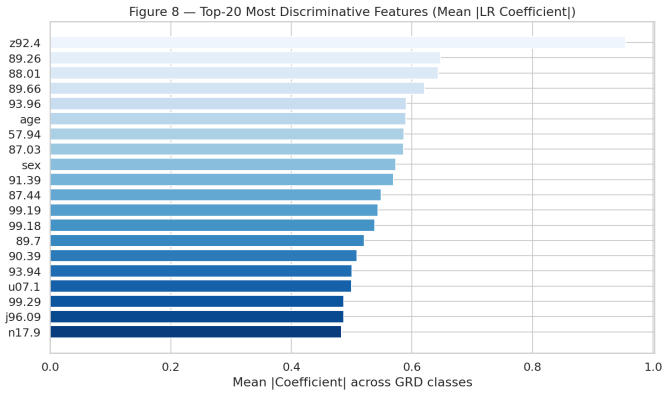

Fig. 9. Top-20 most discriminative features ranked by mean absolute logistic regression coefficient across all GRD classes. CIE-9 procedure codes dominate the importance landscape; CIE-10 diagnostic codes and demographic variables (age = rank 6; sex = rank 9) furnish complementary discrimination for separating obstetric, neonatal, and adult medical GRD profiles.

across all GRD classes. The feature importance landscape is dominated by CIE-9 procedure codes: `89.26` (non-invasive pulse measurement), `88.01` (abdominal CT scan), `87.44` (chest radiography), `57.94` (urinary catheterisation), and `93.96` (oxygen enrichment therapy) emerge as the leading discriminators. Among CIE-10 diagnostic codes, `Z92.4` (personal history of sterilisation), `U07.1` (COVID-19 confirmed), `N17.9` (acute renal failure), and `J96.09` (acute respiratory failure) rank prominently. Patient age (rank 6) and biological sex (rank 9) provide complementary demographic discrimination, particularly for separating neonatal (age ≈ 0), obstetric (age 20–35), and elderly medical GRD profiles.

The MLP's weighted F1-score of 0.8121—6.9 pp above its macro F1 of 0.7433—quantifies the systematic performance gra-

dient between high-frequency and rare GRD classes, reflecting the irreducible consequence of extreme class imbalance. Future work should evaluate cost-sensitive learning and oversampling strategies adapted for binary feature spaces to attenuate this gradient.

G. Comparison with Published Literature

The MLP macro F1-score of 0.7433 on the top-80 GRD subset aligns favourably with international DRG/GRD prediction benchmarks. Janke et al. [9] reported macro F1 of 0.70–0.82 on curated US DRG subsets using logistic regression and random forests; Li et al. [11] achieved accuracy above 0.85 on MIMIC-III via graph-embedding representations. Direct numerical comparison is constrained by differences in label-space cardinality, country-specific grouper logic, and evaluation scope. Critically, our results are obtained exclusively from structured CIE-10 diagnostic and CIE-9 procedural code representations, without recourse to free-text discharge summaries, laboratory values, or clinical narratives—a substantially more constrained information environment than those exploited by leading MIMIC-III studies [8], [10]. Integrating clinical language models or discharge note NLP is anticipated to close the remaining performance gap.

VII. Conclusions

This paper presented a systematic supervised machine learning framework for automated GRD prediction from structured inpatient data at Hospital El Pino, addressing a critical operational need within Chile's FONASA-financed hospital network. Among the four evaluated architectures, the Multilayer Perceptron demonstrates superior discriminative capability, attaining an accuracy of 0.6769 and a macro F1-score of 0.3497 across the full 450-class label space, and an accuracy of **0.8168** with a macro F1-score of **0.7433** on the top-80 GRD subset.

The study establishes that binary TF-IDF multi-hot encoding of CIE-10 diagnostic and CIE-9 procedural code sets, combined with standardised demographic variables, effectively captures a clinically informative 2,995-dimensional representation suitable for high-cardinality multi-class classification. Patient age and biological sex contribute complementary demographic discrimination, particularly for separating obstetric, neonatal, and adult medical GRD profiles. The severe class imbalance (813:1) remains the primary challenge for rare GRD prediction: the 6.9 pp gap between weighted F1 (0.8121) and macro F1 (0.7433) quantifies this effect. The MLP's training efficiency (26 s) further establishes its viability as a low-latency decision-support component in public hospital administrative workflows.

VIII. Limitations and Future Work

Several limitations circumscribe this study. First, the binary TF-IDF encoding treats diagnostic and procedural codes as unordered sets; position-aware or attention-based architectures could exploit the principal-versus-secondary code distinction to improve severity-level discrimination. Second, the current analysis is restricted to structured EHR fields; integrating unstructured discharge summaries via clinical NLP or specialised large language models constitutes a high-priority extension [8]. Third, XGBoost was evaluated under a constrained configuration; a fully hyperparameter-tuned version is expected to be competitive with the MLP. Fourth, advanced class-imbalance mitigation—including cost-sensitive learning, SMOTE-based oversampling for sparse binary spaces, and hierarchical GRD taxonomy classification—warrants systematic evaluation. Fifth, external validity requires multi-centre validation across the Chilean public hospital network prior to clinical deployment. Finally, Explainable AI (XAI) techniques such as SHAP or LIME must be integrated to furnish interpretable per-prediction justifications, a prerequisite for building clinician trust in automated coding recommendations.

IX. Data and Code Availability

The complete source code for preprocessing, exploratory data analysis, model training, and evaluation, together with the final trained MLP model and all generated figures, is publicly available at:

[https://github.com/Cristxpherr/CINF104-GRD-HospitalElPino]

The repository contains: (i) the end-to-end Python pipeline (`grd_pipeline.py`); (ii) the serialised MLP model; (iii) all nine EDA and experimental figures; and (iv) a `README.md` detailing the steps required to reproduce all results from the raw `dataset_elpino.csv` file. The clinical dataset constitutes identifiable patient health information and is therefore not publicly redistributable; researchers seeking to replicate this study with the original data should contact the Hospital El Pino research office directly. All code is released under the MIT License.

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